

# Baochen Tian

Research Assistant, Australian National University

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## EDUCATION

### Australian National University

Graduated 2025

**Bachelor of Science in Physics** | GPA: 6.13 / 7.0

Attended 2023 - 2025

Exchange: National Taiwan University, Spring 2025

### Beijing Institute of Technology

2020 - 2023

Undergraduate studies; transferred to ANU in 2023

International Economics and Trade (2020 - 2021); Optical-Electronics Information Science (2021 - 2023)

## INDEPENDENT PROJECTS

### Flow Matching Reimplementation of DiffPhyCon

2026

Derived and implemented a flow-matching formulation of the joint trajectory model in [DiffPhyCon](#) (NeurIPS 2024), a diffusion-based framework for physical system control. Matched the original model capacity and training configuration. Compared reconstruction and control against DDPM/DDIM baselines on 1D Burgers and 2D Jellyfish.

Investigated guidance and prior reweighting under flow matching, including behavior unexplained by a naive score-to-velocity conversion and the need for an FM-specific derivation.

Code: [github.com/bcTiann/diffphycon](https://github.com/bcTiann/diffphycon)

## RESEARCH EXPERIENCE

### Python Pipeline for Post-processing QUOKKA Simulations

July 2025 - Present

Research School of Astronomy and Astrophysics, ANU.

Research Assistant (April 2026 - Present). Supervisor: Dr. ChongChong He.

Investigating how reliably emission tracers (CO, H-alpha, C+, H I) recover the mass-weighted velocity dispersion in simulations, and how this depends on gas phase.

Developing a Python pipeline that converts 3D [QUOKKA](#) snapshots into synthetic emission-line observations with [DESPOTIC](#), [Grackle](#) and [Cloudy](#).

Code: [github.com/bcTiann/quokka\\_postprocess\\_3D](https://github.com/bcTiann/quokka_postprocess_3D)

### Deep Learning for CMB Inpainting

February - June 2025

LeCosPA, National Taiwan University, during an exchange semester.

Supervisor: Prof. Pisin Chen; mentor: Reyhan D. Lambaga.

Applied Transformers and DDPMs to inpaint CMB maps masked by the galactic plane. Evaluated reconstruction fidelity by comparing the angular power spectra of generated maps against ground truth.

### Rectangular Haloscope for Axion Detection

January - February 2024

Quantum Technologies and Dark Matter Research Lab, University of Western Australia.

Supervisors: Prof. Michael Tobar and Dr. Aaron Quiskamp.

Modelled a tunable-wall rectangular haloscope in COMSOL, simulating resonant modes and evaluating sensitivity against cylindrical tuning-rod geometries.

## SCHOLARSHIPS & SKILLS

### EQUUS Summer Research Scholarship

January - February 2024

Six-week research program at the University of Western Australia, supported by the ARC Centre of Excellence for Engineered Quantum Systems.

**Machine learning:** PyTorch, Transformers, VAEs, diffusion models and flow matching.

**Astrophysical modelling:** [QUOKKA](#) post-processing, synthetic emission-line observations, [DESPOTIC](#), [Grackle](#) and [Cloudy](#).

## OTHER INTERESTS

**Chamber music and performance:** Founded chamber ensembles at BIT and ANU; pianist for the BIT University Choir; participated in two university music competitions and 7+ community performances. Performed in NTU's 2025 Symphony Orchestra Chamber Music Night and was invited to perform at an official ANU welcome dinner for visiting scholars.